

c00gle

112306005 吳沛霓 112306034 潘振軒 112306039 陳睿哲
112306071 陳厚銘 112306073 李岳融



OUR PROJECT : GAMES

- Each of our team members are game lovers
- Google results might not be relevant to the topic "game"
- Google might confuse A with B (especially common nouns)



VALORANT



LEAGUE^{OF}
LEGENDS

JAVA CLASSES

Frontend

coogle.html
result.html*

Application Layer

DemoApplication

Controller Layer

GoogleController
SuggestionController*

Service Layer

GoogleService*
Keyword
KeywordList*
PageComparator
PageHeap*
WebPage*
WebTree*
WordCounter*

ADDITIONAL FUNCTIONS

Controller Layer

- SuggestionController : autocomplete suggestions

Service Layer

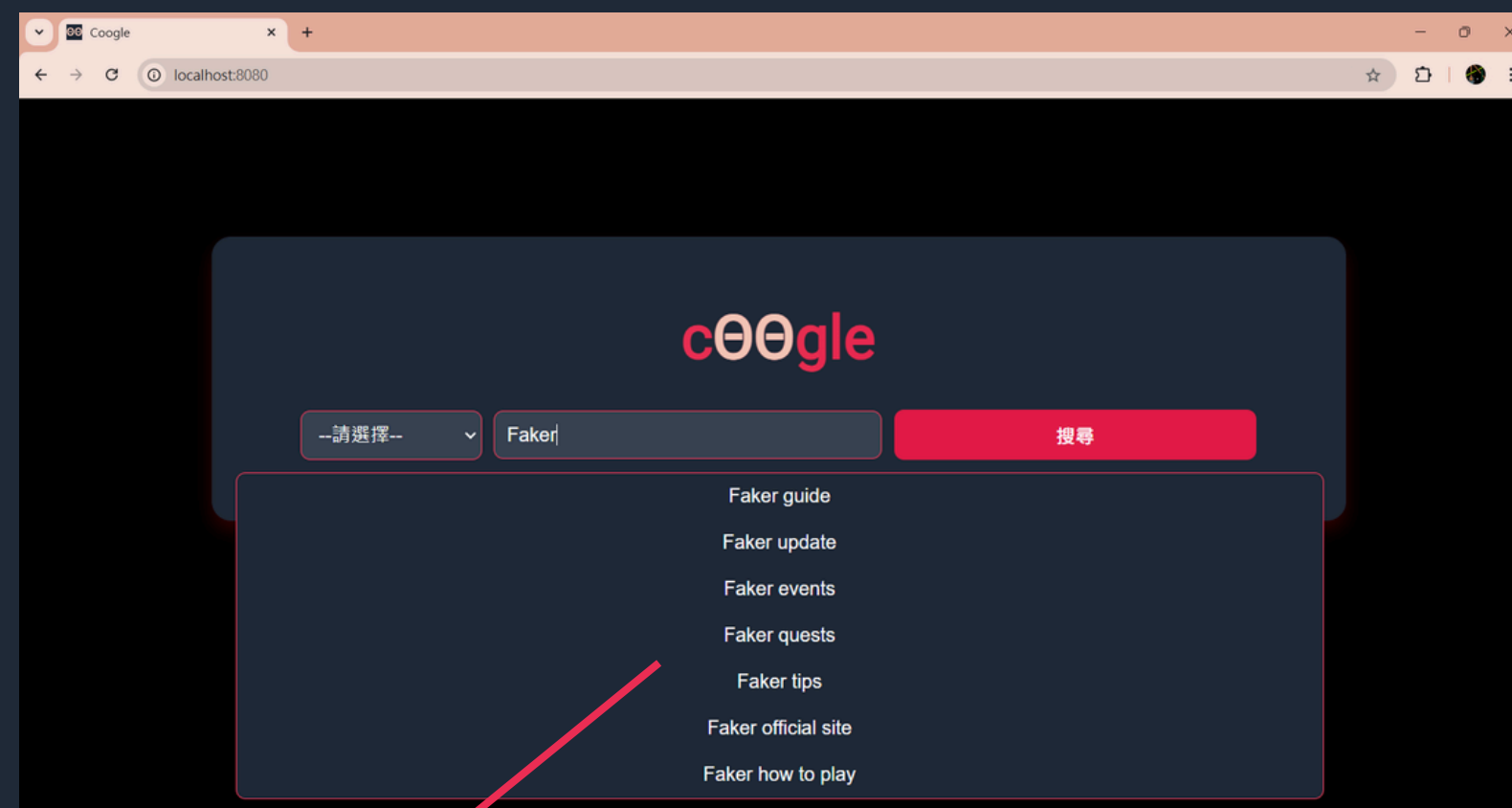
- GoogleService : add filter
- KeywordList : switch TXT file with filter
- PageHeap : Heap(PriorityQueue)
- WebPage : check if there are any subpages
- WebTree : create tree for subpages (post order)
- WordCounter : GZIP/ Boyer-Moore



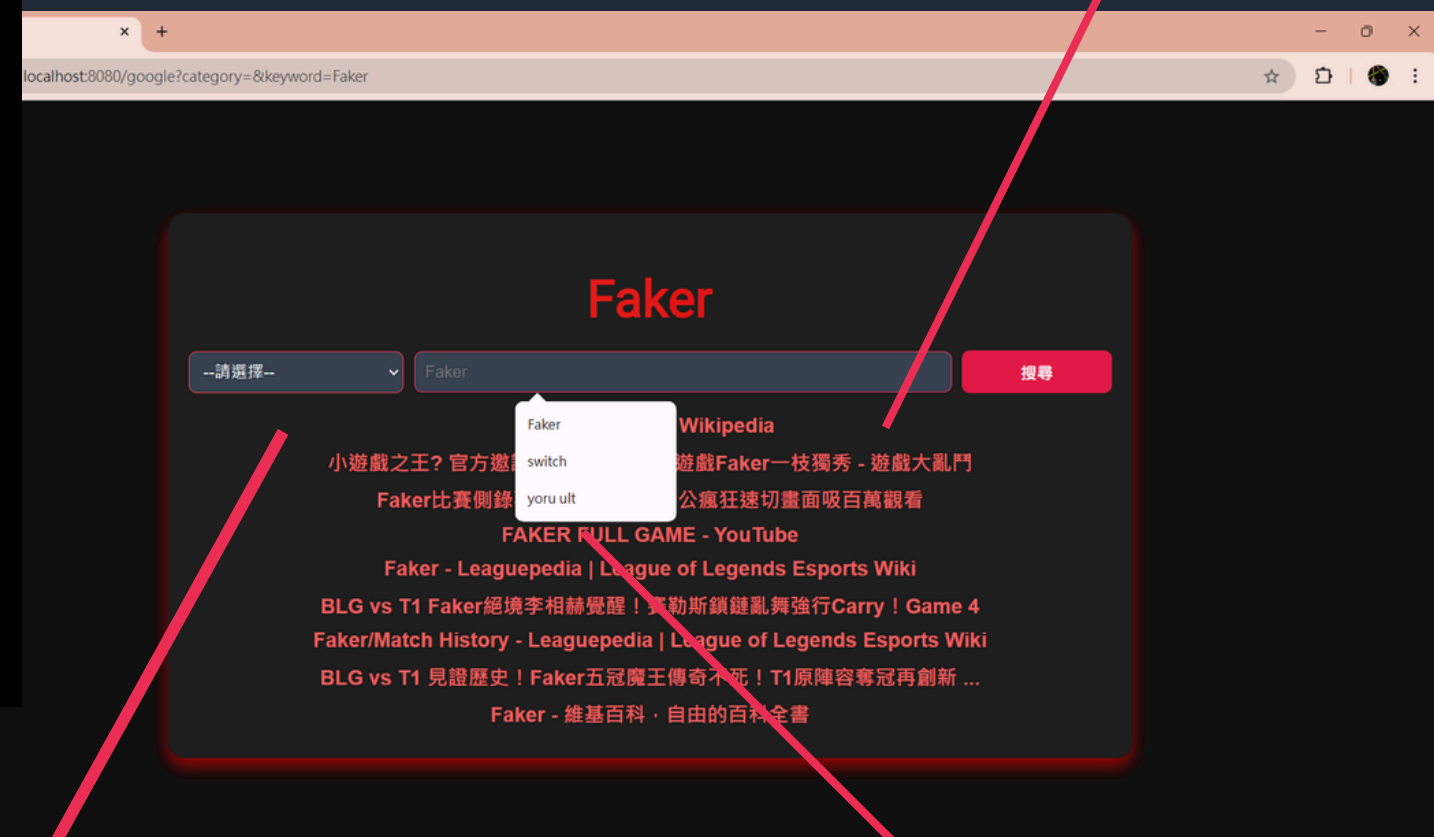
OUR PAGE

- Esport style webpage

- Search result



- Autocomplete suggestions



- filter

- Search history



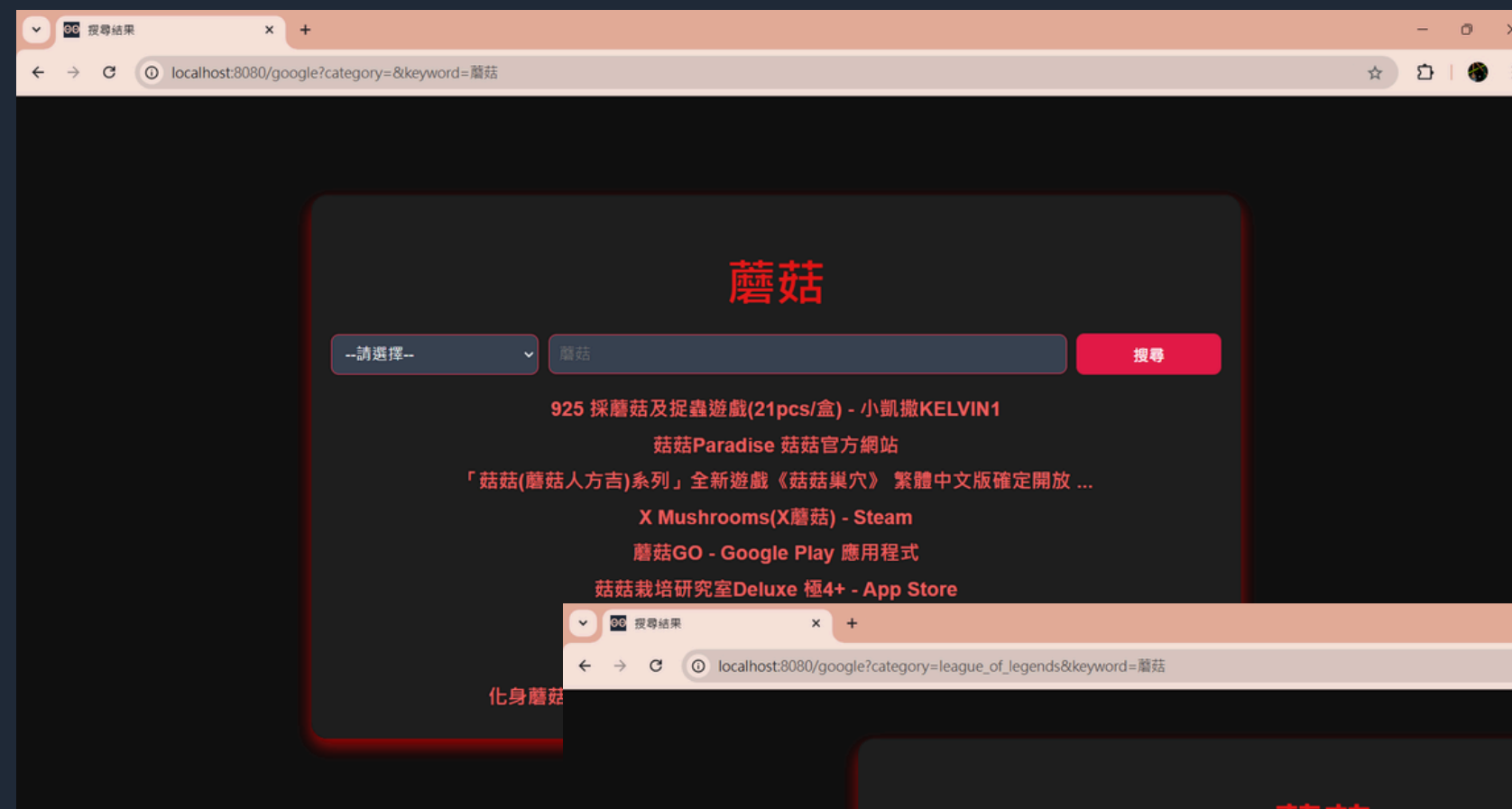


FILTER

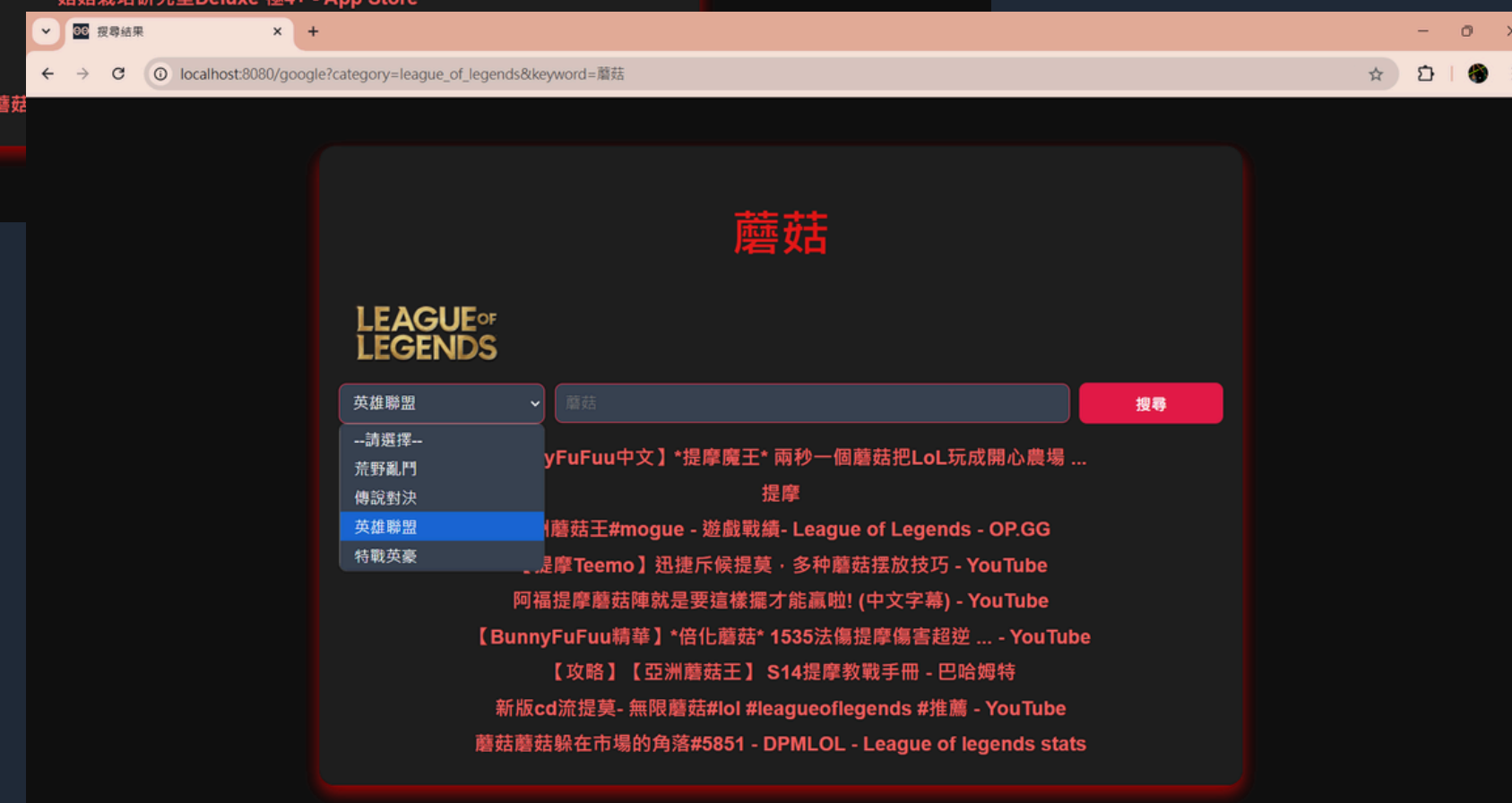
Search more directly

We created a filter with a drop-down menu for users to make sure the results are relevant to the game they are looking for.

The methods in the GoogleService class and KeywordList class will adjust according to the options selected by the user in the drop-down menu.



(without filter)



(WITH filter)

PROBLEMS & SOLVEMENT

efficiency issues (code from WordCounter class)

- The Boyer-Moore Algorithm : HW3
- GZIP : compress the content before sending it

```
29● public String fetchContent() throws IOException {
30     URL url = new URL(this.urlStr);
31     HttpURLConnection conn = (HttpURLConnection) url.openConnection();
32     conn.setRequestProperty("Accept-Encoding", "gzip");
33
34     InputStream in = "gzip".equals(conn.getContentEncoding())
35         ? new GZIPInputStream(conn.getInputStream())
36         : conn.getInputStream();
37
38     try (BufferedReader br = new BufferedReader(new InputStreamReader(in))) {
39         StringBuilder retVal = new StringBuilder();
40         String line;
41         while ((line = br.readLine()) != null) {
42             retVal.append(line.toUpperCase()).append("\n");
43         }
44         return retVal.toString();
45     }
46 }
```

```
91● public int BoyerMoore(String T, String P) {
92     int n = T.length();
93     int m = P.length();
94
95     int i = m - 1;
96     int j = m - 1;
97     int count = 0;
98
99     while (i < n) {
100         if (T.charAt(i) == P.charAt(j)) {
101             if (j == 0) {
102                 count++;
103                 i += m;
104                 j = m - 1;
105             } else {
106                 i--;
107                 j--;
108             }
109         } else {
110             int l = last(T.charAt(i), P);
111             i += m - Math.min(j, l + 1);
112             j = m - 1;
113         }
114     }
115     return count;
116 }
117 // http://soslab.nccu.edu.tw/Welcome.html
118
119● public int last(char c, String P){
120     for(int i = P.length() - 1 ; i >= 0 ; i--) {
121         if(P.charAt(i) == c) {
122             return i;
123         }
124     }
125     return -1;
126 }
```



**THANK
YOU**

